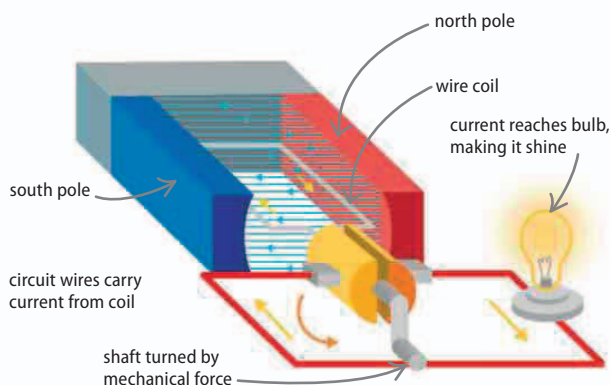


Direct current generator

Generators can be built to produce either direct current (DC) or alternating current (AC) (see page 216). A DC generator has the same parts as a DC electric motor but works in reverse (see page 212). The wire coil of the conducting wire is turned inside a magnetic field that is generated by a large permanent magnet. As the wire in the coil cuts across the magnetic field lines, a voltage and current are created in the coil.



△ Stage 1

An experimental direct current generator sees the wire coil turned by a hand crank. As it passes through the magnetic field of the permanent magnet, a current is induced in the wire coil.

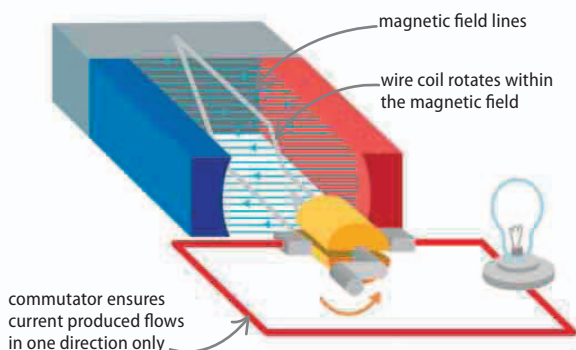
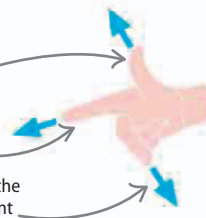
▷ Right-hand rule

This rule shows the direction in which a current will flow in a wire when the wire moves in a magnetic field.

thumb shows the direction of thrust applied to the wire

first finger points in the direction of magnetic field

middle finger shows the direction of the current

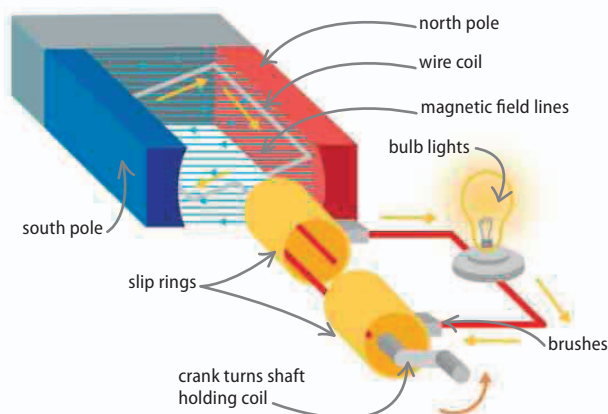


△ Stage 2

An electric current is only generated when the wire coil is cutting the horizontal magnetic field lines. When the wire coil is vertical, no current is produced and the bulb does not light up.

Alternating current generator

An alternating current (AC) generator, known as an alternator, does not use a commutator. As a result, the current produced changes direction twice for every complete 360° turn of the coil. Individual slip rings are fitted to each of the two ends of the coil to provide a path for the current to leave. Brushes contact the slip rings and complete the path for the current into the circuit to which the generator is attached.



◁ Alternator in action

This simple alternator has a single loop of wire acting as the coil. Turning a hand crank rotates the coil between the poles of a permanent magnet. An alternating current is induced in the coil, which flows through the slip rings and brushes to light the bulb.

REAL WORLD

Wind-up electrics

In parts of the world where electricity is unreliable or absent, and batteries are expensive, radios (as below), laptops, and other electronic devices can be powered by hand. A small generator inside the device is turned by a hand crank to charge up the rechargeable batteries inside.

